بسم الله الرَّحمن الرَّحیم

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1- Design a CNN network for CIFAR-10 classification.
    a. Train the designed model with at least three different batch sizes and three different optimizers.
    b. Select the best model using validation data.
    c. Visualize the training process and analyze the results using appropriate methods, graphs, and evaluation metrics.
    d. Evaluate the designed and trained model with MNIST test data, compare the results with CIFAR-10 test data, and analyze the outcomes. If there is a need, apply an appropriate transformer to the inputs.
    e. Train the model using MNIST training data, evaluate and analyze the results, and compare them with previous results.

2- Design a CNN network for the Intel Image Classification dataset.
    a. Train the model and find the best weights.
    b. Evaluate the model on the test data.
    c. Attempt to find a better model using transfer learning techniques.
    d. Analyze and report the results.
    e. Use the best model as a feature extractor and extract the training, validation, and test features from the second-to-last layer.
    f. Create three folders for the training, validation, and test data.
    g. Save the features of each image as a NumPy array in the corresponding folder, ensuring that the filename allows retrieval of both the name and the class of the image.

موفق باشید.